Cycle In Graph

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The problem

- Directed graph as adjacency list; does it contain a cycle?
  3-colour DFS.

Intuition

- A cycle exists if depth-first search meets a node that's still on the current recursion stack.

Approach

- 1. Colour nodes white, grey, black
- 2. Grey means on the stack
- 3. Edge to a grey node is a back edge, a cycle

Java solution

```java
public class CycleInGraph {
    public static boolean cycleInGraph(int[][] edges) {
        int[] color = new int[edges.length];
        for (int v = 0; v < edges.length; v++) if (color[v] == 0 && dfs(v, edges, color)) return true;
        return false;
    }

    private static boolean dfs(int v, int[][] edges, int[] color) {
        color[v] = 1;
        for (int w : edges[v]) {
            if (color[w] == 1) return true;
            if (color[w] == 0 && dfs(w, edges, color)) return true;
        }
        color[v] = 2;
        return false;
    }
}
```

Complexity

- Time: O(v + e)
- Space: O(v)

Recap

- A cycle exists if depth-first search meets a node that's still on the current recursion stack.
- Complexity: O(v + e)